

Analyzing training needs and operational tradeoffs at the St. James Food Pantry: a case study

Leonardo Marciaga

Abstract

Food insecurity is a major problem in the US, with 13.5% of households facing a food-insecure status as of 2023 (United States Department of Agriculture [USDA], 2024), which stresses the critical role that food pantries play in addressing this issue. One particular challenge concerning food pantry operations is that of training volunteers: as these are non-profit organizations, they are often resource-constrained in terms of training infrastructure, yet these resources are necessary to guarantee their sustainability. We focus on the training process for episodic volunteers at the St. James Food Pantry, located in Chicago. This process presents several challenges, such as identifying specific training components, constraints in the trainer pool, and assignment and time considerations. In this work, we propose a framework to address these challenges. First, we identify relevant training components, tasks, and roles via a training needs assessment procedure. Then, we address trainer constraints, assignment and time considerations via a constraint programming model embedded in a decision support tool that maximizes total assigned preference and minimizes makespan. We compare its performance to two simpler lexicographic models that prioritize either of those metrics. Our model achieves higher total assigned preference than the makespan-first baseline (8.76% on average) with no increase in tardiness (delay in training completion). Additionally, under tighter trainer constraints, the model matches makespan-first behavior, while under looser constraints, it matches preference-first behavior.

1 Introduction

Food insecurity is a major problem in the US, with 13.5% of households facing a food-insecure status as of 2023 (USDA, 2024) and, more specifically, 25% of people in the greater Chicagoland area are facing food insecurity (Greater Chicago Food Depository et al. [GCFD et al.], 2025). The labor of food pantries is therefore crucial to mitigate this issue, and they have played this role for over four decades (Esmaeilidouki et al., 2023). Their staff is typically formed by volunteers (Esmaeilidouki et al., 2023), and since food pantries are nonprofit organizations and as such are resource-constrained, they need an appropriate amount of volunteer staff in order to fulfill their mission (Berenguer & Shen, 2020; Kaur et al., 2022). Therefore, food pantries require avenues to recruit and train new volunteers to ensure the continuity of their operations.

Volunteers generally fall into two categories: *formal* and *episodic* (Hustinx et al., 2008; Rehnberg & Moore, 2011). Formal volunteers, also known as *ongoing*, show a long-term commitment with the organization and dedication to the cause, often resembling paid staff due to the regularity of

their service. On the other hand, episodic volunteering is defined as service of short duration, either a one-time basis or a specific project that recurs each year (Macduff, 1988). These make up the majority of the volunteer workforce (Cnaan et al., 2022), participate intermittently and are more drawn to short-term tasks (Hustinx & Lammertyn, 2003). Their recruitment also imposes some challenges, like additional preparation in advance and time commitment from management staff (Rehnborg & Moore, 2011). Still, there exist opportunities in bringing in episodic volunteers into a nonprofit’s operations: aside from being the more available type of volunteers and thus more easily recruitable, the organization can secure assistance not available otherwise, and these volunteers can help promote it (Rehnborg & Moore, 2011). Additionally, volunteers’ experience, if rewarding, can inspire them to become future donors (Wu et al., 2024).

Training is also a crucial part of nonprofits’ operations, since it provides the foundation for the volunteers’ relationship with the organization (Hood, 2011). We may distinguish between *orientation*, which consists of a broad overview of the organization, its mission, culture, norms, and behavioral expectations, while *training* itself provides volunteers with the tools to successfully execute their tasks. Crucial to designing a training program is the needs assessment process (Goldstein, 1993), which considers a wide range of organizational aspects to inform the training program. In fact, Macduff (1988) emphasizes the importance of the needs assessment process as part of a strategy to ensure the success of episodic volunteers. Still, training has been long recognized to be an important constraint for nonprofits (Cunningham, 1999), since these are limited in resources and may lack infrastructure components for training, yet these resources are necessary to ensure the sustainability of the organization.

This work focuses on the training process for episodic volunteers at the St. James Food Pantry, located in Chicago. More specifically, we perform a needs assessment process for episodic volunteers assisting with the pantry’s regular on-site distribution program and identify important training components that may help improve the pantry’s training program. Additionally, we address the operational challenge of training by considering a constraint programming model embedded in a decision support tool that aids with creating volunteer role assignments and training schedules subject to staff, space, and time constraints. Section 2 provides an overview of the pantry’s operations. Section 3 details the steps of the needs assessment process in the context of the pantry. Section 4 details the formulation of the mathematical model, while Section 5 contains the model’s evaluation methodology and results. Section 6 includes a discussion of our findings and potential avenues for future work.

2 St. James overview

The St. James Food Pantry is located in the city of Chicago at 2907 South Wabash Avenue, and it is affiliated with the nearby St. James Catholic Church. It serves an average of 1500 households per month in Chicago’s Bronzeville area (St. James Catholic Church, n.d.-a), offering both on-site and off-site distribution programs, as well as external services via social service partnerships (St. James Catholic Church, n.d.-b). Its operations are led by the pantry’s director, while specific tasks related to volunteer management are the responsibility of the volunteer coordinator. Both the director and the volunteer coordinator oversee the training process for new volunteers.

Regular volunteers are mostly parish members from the St. James Church as well as local community members from the Bronzeville area. Prospective new volunteers may use the pantry’s online

registration form, as well as directly contacting the pantry leadership staff. Particularly, organizations such as high schools, colleges, or companies may contact the pantry leadership staff for community service opportunities, allowing a group of people from the organization to volunteer for the pantry for a certain number of days. We will be concerned with the training process for new volunteers recruited in this way, since these tend to form a large group (relative to the pantry’s regular staff and physical capacity) that must be trained at once.

We focus on the regular on-site distribution process, for which the pantry follows a client-choice model, in which clients (i.e., people requesting the pantry’s services) are guided by volunteers and pick their items, akin to the experience of shopping in a grocery store. We will borrow the term “shopping” to refer to clients picking items in this way. Figure 1 shows an overview of the on-site distribution process. In particular, we make a distinction between “orientation and setup process training” and “shopping process training.” When arriving, new volunteers receive orientation about the pantry’s general mission and values. New volunteers may then be assigned tasks on the food setup process, receiving training for them along the way. Afterwards, all new volunteers, regardless of the task they were doing before (if any), receive training for the subsequent shopping process, in which clients are allowed access to the pantry’s facilities.

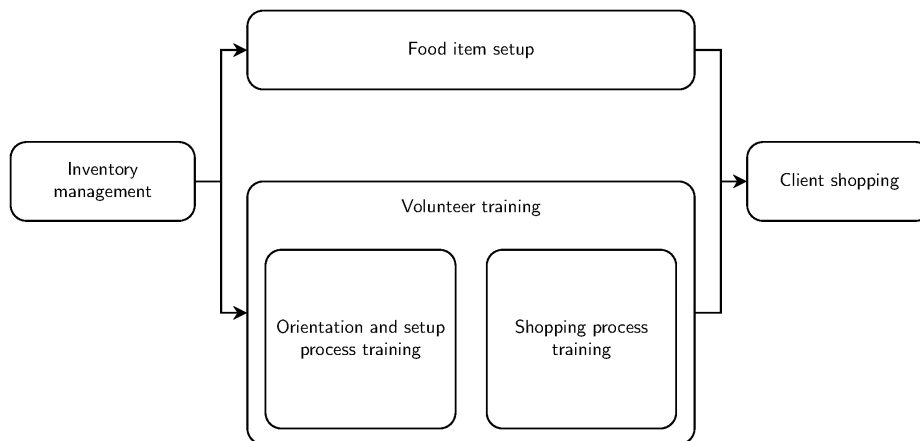


Figure 1: Process map of the St. James Food Pantry’s regular on-site distribution process. Closing-hour procedures, administrative tasks, and other programs are not included in this diagram.

Among other challenges, the pantry faces space, staff, and time constraints to carry out this process. The pantry’s physical space supports its regular staff and client intake, and therefore the arrival of relatively large groups of new volunteers may introduce difficulties. Also, while other staff may assist with the shopping process training, the director and volunteer coordinator (which may not both be available on the same day) are the ones responsible for the orientation and setup process training, in which volunteers may be assigned different tasks depending on daily needs. Additionally, the pantry has specific opening hours at which clients are allowed access to the pantry’s facilities for the shopping process, and this time may be difficult to meet with large groups of new volunteers due to uncertainty in their arrival time, as well as the aforementioned staff and space constraints. In particular, in the setup process training, allowing a greater diversity of tasks that volunteers execute may impose demands on the staff overseeing this training and the total time that this

process takes, potentially running over the specified opening hours.

We thus develop a needs assessment procedure to aid the pantry staff, both the ones currently in charge of the training process and other staff that may later serve as trainers. Given that volunteer-to-task assignment on the setup process training may depend on daily needs, we also develop a mathematical model of this subprocess to study the space, staff, and time constraints occurring in this process.

3 Training needs assessment

In this section we perform a training needs assessment to identify important training components for episodic volunteers. This is a procedure that takes into account tasks, behaviors and environment within an organization to inform the training process (Goldstein, 1993). Even though a training process is already in place the pantry, we will see that there exist a number of considerations that need to be communicated to volunteers. It becomes difficult to communicate these considerations especially in situations with low regular volunteer staff and/or a high number of new, episodic volunteers relative to the available regular volunteer staff, and thus we perform this assessment to identify important components that must be communicated to volunteers and provide a guide to the regular volunteer staff regardless of the number of new episodic volunteers requiring training.

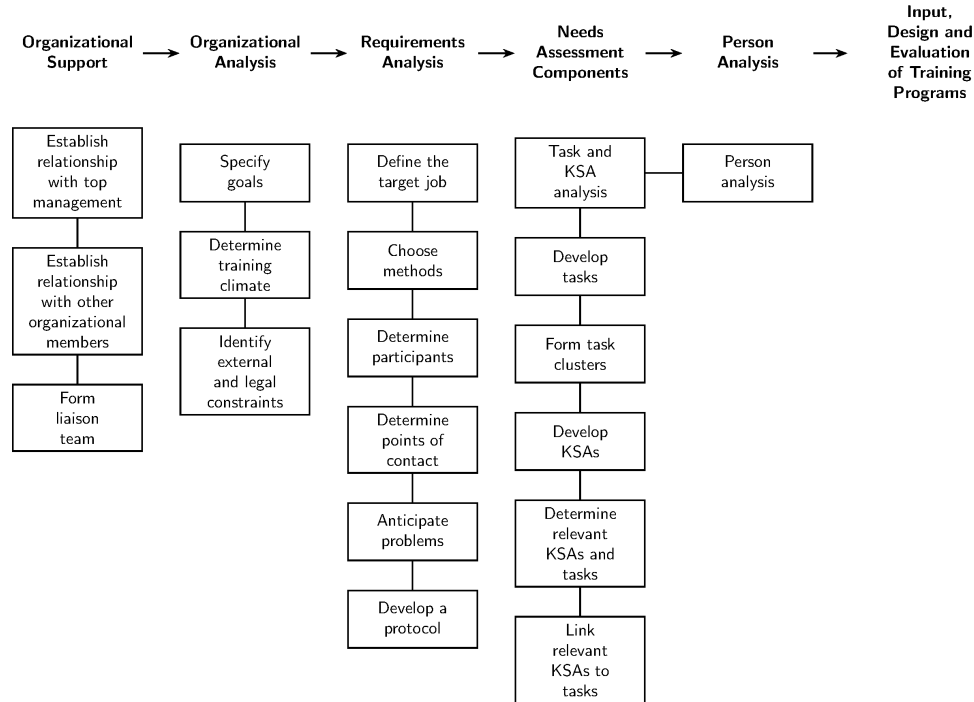


Figure 2: Model of the needs assessment procedure (Goldstein, 1993).

Central to the training needs assessment procedure is the identification of relevant tasks that need to be performed and knowledge, skills, and abilities (KSAs) that are needed to perform this tasks. Goldstein (1993) mentions the definition of these categories due to Prien:

Knowledge (K) is the foundation of skills and abilities, and it refers to the background information which, when applied, allows for performing a task successfully.

Skill (S) is the capacity to perform job functions, and they often refer to psychomotor-type activities.

Ability (A) usually refers to cognitive capabilities necessary to perform a job function.

We followed the model outlined in Figure 2 to perform the needs assessment procedure. We secured organizational support from the pantry’s director, the main point of contact for this analysis, as well as the volunteer coordinator. In what follows, we detail our organizational analysis, requirements analysis, and needs assessment components. Observe that person analysis is not quite applicable in this context since we are concerning ourselves with new episodic volunteers, which by definition are new to the pantry’s operations and thus require training regardless.

3.1 Organizational analysis

In this step of the assessment, we consider three aspects: specification of organizational goals, determination of training climate, and identification of external and legal constraints. The pantry’s main operational goal is to help its clients meet their basic needs and live with dignity (St. James Catholic Church, [n.d.-a](#)). The pantry requires volunteers to help run its programs, particularly, the on-site food distribution. Aside from ensuring the continued operation of the pantry, there is a focus of developing social responsibility within its volunteers. The pantry has partnerships with organizations, such as high schools, universities, or companies. It receives groups of episodic volunteers from these organizations that assist with operations for a certain time, in addition to its regular volunteer staff. The pantry then aims to instill the sense of social responsibility in these groups through direct service, typically in the on-site distribution process detailed in Section 2.

Determining the training climate consists of examining the organization’s components that may affect a recently arrived trainee (Goldstein, 1993). Since we are limiting ourselves to the episodic volunteers that assist with the on-site distribution process, this step consists of determining what organizational aspects may support or hinder the newly trained episodic volunteers. We identify that training imparted by the director/volunteer coordinator themselves as well as support from the director, volunteer coordinator, and regular volunteer staff supports the training climate, while regular staff constraints constitutes a challenge to it. This last consideration entails that it may be difficult to supervise volunteers and provide feedback on days with few regular staff and many new, episodic volunteers relative to the regular staff available.

External and legal constraints are in general handled by the director and volunteer coordinator directly. The pantry belongs to the partner agency network of the Greater Chicago Food Depository, which maintains a series of guidelines for its partner agencies (Greater Chicago Food Depository [GCFD], [n.d.](#)), in particular concerning food storage, distribution, delivery, and client intake methods (GCFD, 2025). Other agencies’ guidelines also condition the pantry’s operation, such as the United States Department of Agriculture (USDA), the Federal Emergency Management Agency (FEMA), and the Department of Homeland Security (DHS). Additionally, the Archdiocese of Chicago is an important stakeholder in the pantry’s operations, given the pantry’s ties to the St. James Catholic Church. While the director and volunteer coordinator are the ones responsible for overseeing these regulations and not the newly arrived episodic volunteers, there are both safety and client-related considerations that volunteers need to receive instruction about, which may leave the pantry at the risk of lawsuits. These considerations will be mentioned in more detail at the end of Section 3.3.

Process	Role	General description
Setup	Non-physically intensive setup	Arranges food items for the shopping process. May perform other minor tasks depending on need.
	Physically intensive setup	Transports heavy items from inventory locations.
Shopping	Shopping aide	Guides clients through the pantry’s shelves to help them select their food items.
	Distribution aide	Distributes meat, eggs, dairy and/or frozen products in a section of the shopping process.

Table 1: Roles that episodic volunteers typically perform during the on-site distribution process. It is important to note that here are heavy loads to be carried by the volunteers during the setup process (e.g., food boxes and bags) that inherently demand physical ability, and thus such a task cannot necessarily be assigned to any volunteer in the volunteer pool.

3.2 Requirements analysis

The requirements analysis is an examination of the details that must be cleared out to guarantee the success of the needs assessment procedure (Goldstein, 1993). We follow the steps outlined in Figure 2.

The target job, which in this case corresponds to the labor of a volunteer, varies according to the task that volunteers are assigned to. In this case, we can identify roughly four different types of roles that episodic volunteers may be performing during the on-site distribution process: two during the setup process, and two during the shopping process. These roles are outlined in Table 1. It is important to note that here are heavy loads to be carried by the volunteers during the setup process (e.g., food boxes and bags) that inherently demand physical ability, and thus such a task cannot necessarily be assigned to any volunteer in the volunteer pool. Therefore, only those meeting the physical ability requirements may perform such tasks, and they tend to perform this task for as much time as necessary. Those not meeting the requirements are assigned other tasks in the setup process depending on availability.

Methods chosen to perform the needs assessment were observation and questionnaires administered to the director. The author performed on-site observations at least once a week throughout the scope of the project (a semester, i.e., 16 weeks) to minimize interruption of routine work, which was achieved via general organizational support. These observations were used to develop task statements and KSAs for each role. Additionally, questionnaires to determine relevant tasks and KSAs were administered to the director, whose questions are included in the Appendix. Regular meetings were held with the director as well for clarification and additional questions.

The director was the only staff member and point of contact involved in this procedure, given their responsibility for the training procedure. While it would be useful to include episodic volunteers themselves, there are few ways to communicate with them before or after their service, making it difficult to obtain feedback from them. One issue that may have affected the assessment were ongoing regular staff changes. The volunteer coordinator would have been the next immediate point of contact to reach out to, but throughout the scope of the project, the director point of contact remained stable.

1	Not important
2	Slightly important
3	Important
4	Very important
5	Crucial

Table 2: Importance scale for tasks and KSAs.

3.3 Relevant tasks and KSAs

We report the most relevant tasks and KSAs for each of the roles recognized in Table 1, as well as which KSAs are important for which tasks. Questionnaires were administered to the pantry director, in which they were asked to rate the importance of the developed tasks and KSAs using the scale in Table 2. Here we report the task and KSAs that were rated a 4 or higher.

Developed tasks and KSAs for each role, as well as the rated importance for each, are reported in the Appendix. These results reveal that all tasks save for two in the physically intensive setup role (indicated in Table 13) were rated 4 or higher. KSAs that were rated 4 or higher are indicated in Table 19 in the Appendix.

Table 3, Table 4, Table 5, and Table 6 contain the relevant tasks for each role, as well as the most relevant KSAs for each task. Each KSA was chosen from Table 19 on the basis of whether it is at least helpful in performing the corresponding task. An exception to this rule is the last task from Table 6, which is linked to the KSA “Skill in opening boxes using sharp tools” with importance 3.5 for its role (see Table 18).

KSAs with particular importance (rated more than 4 in importance) are those related to item limit guidelines and operation of the pantry’s shopping and storage carts. Item limit guidelines are crucial to manage the pantry’s inventory, and thus remain an important component that must be communicated to volunteers. Shopping and storage cart operation acquires such importance specifically because of the pantry’s shopping cart procedure during the shopping process. There exist different rules depending on whether a client has their own cart or whether they require the pantry’s shopping carts instead, which constitute different procedures. If not followed, accidental damage of the pantry’s cart or, even worse, the client’s, can be costly at best and lead to legal issues at worst.

Other KSAs worth mentioning are physical lifting/sharp tool operation and cross-language communication. Volunteers may be at risk of physical injuries due to improper lifting technique or sharp tool mishandling, and thus ensuring proper safety in the training procedure remains important. Additionally, the pantry receives clients with limited English proficiency, and thus volunteers must at least be aware of such considerations to guarantee a proper service.

4 Mathematical model

We want to study the assignment of large groups (relative to the pantry’s regular staff) of episodic volunteers to certain roles during the setup process, considering time and physical space constraints. However, keeping track of the total process duration is not trivial, given that some parts of the

Task	KSA
Arrange chairs, tables, shopping carts, and/or sanitizing items in the pantry’s main room for client intake	Skill in moving light chairs or tables (up to 10 lb) Skill in placing items in plastic bags or bins Ability to understand verbal and visual instructions for daily tasks Ability to communicate task progress or issues to the director or other volunteers
Unpack and place food items in designated bins, following the pantry’s guidelines	Skill in opening boxes using sharp tools Physically intensive setup Ability to understand verbal and visual instructions for daily tasks Ability to communicate task progress or issues to the director or other volunteers

Table 3: Relevant KSA-task links for the non-physically intensive setup role.

Task	KSA
Load, transport, and unload item boxes/bags from storage to food stations or vice versa, using the pantry’s designated carts.	Knowledge of proper lifting technique Skill in lifting boxes, bags, chairs, and tables of up to 50 lb of weight Skill in opening boxes using sharp tools

Table 4: Relevant KSA-task links for the physically intensive setup role.

Task	KSA
Guides clients through the pantry facility to provide information about food item options	Knowledge of the pantry’s item limit guidelines
Aids clients in carrying their chosen food items using the pantry’s shopping carts	Skill in operating the pantry’s shopping and storage carts
Communicates item limit rules to clients using the facility’s visual guidelines	Ability to use visual guidelines to communicate with clients speaking in a different language than the volunteer’s

Table 5: Relevant KSA-task links for the shopping aide role.

Task	KSA
Distributes the day's meat, dairy, eggs, and/or frozen items to clients following item limit rules	Knowledge of the pantry's item limit guidelines Ability to understand visual guidelines for item limit rules
Communicates with shopping volunteers to determine appropriate item limit rules to follow for each specific client	Knowledge of the pantry's item limit guidelines Ability to understand visual guidelines for item limit rules
Replenishes item bins with boxes from storage provided by other volunteers	Skill in opening boxes using sharp tools

Table 6: Relevant KSA-task links for the distribution aide role.

setup process occur in series (such as training volunteers in one role and then training volunteers in another role) and some in parallel (such as volunteers working at the same time), and the extent to which this is the case depends on the number of available trainers. In this section, we achieve these goals via a constraint programming (CP) model that captures organizational preferences, limitations in the number of trainers, as well as process duration. We compare this model to two other models that prioritize only organizational preferences or process duration, respectively.

4.1 Preliminaries

This mathematical preliminaries subsection has three aims: first, introducing the necessary concepts motivating our model formulation; second, identifying prior work in the literature regarding mathematical models for volunteer assignment; third, introducing and reviewing CP formulations of scheduling models. Of particular importance are Section 4.1.3 and Section 4.1.4, which provide the most information on our modeling approach.

4.1.1 Generalized assignment problem

Given matrices $\mathbf{A}$, $\mathbf{B}$, and vectors $\mathbf{b}$, $\mathbf{c}$, $\mathbf{d}$, the problem

$$\begin{aligned}
\min \quad & \mathbf{c}^T \mathbf{x} + \mathbf{d}^T \mathbf{y} \\
\text{s.t.} \quad & \mathbf{Ax} + \mathbf{By} = \mathbf{b} \\
& \mathbf{x}, \mathbf{y} \geq \mathbf{0} \\
& \mathbf{x} \text{ integer,}
\end{aligned}$$

where $\mathbf{x}, \mathbf{y}$ are the variables, the function

$$\mathbf{c}^T \mathbf{x} + \mathbf{d}^T \mathbf{y}$$

is the *objective function*, and the identities

$$\mathbf{Ax} + \mathbf{By} = \mathbf{b}$$

$$\mathbf{x}, \mathbf{y} \geq \mathbf{0}$$

$$\mathbf{x} \text{ integer,}$$

are the *constraints*, is called the *mixed integer programming* problem (Bertsimas & Tsitsiklis, 1997). If there are no continuous variables $\mathbf{y}$, the problem is called the *integer programming* problem. Many problems in the field of discrete optimization can be modeled as integer programs.

One such problem is the *generalized assignment problem* (GAP), which seeks to find the minimum cost (or maximum value) assignment of tasks to individuals subject to capacity constraints (Kundakcioglu & Alizamir, 2009). Given n tasks and m individuals, the problem is formulated as follows:

$$\min \quad \sum_{i=1}^m \sum_{j=1}^n c_{ij} x_{ij} \tag{1}$$

$$\text{s.t.} \quad \sum_{j=1}^n a_{ij} x_{ij} \leq b_i \quad \forall 1 \leq i \leq m \tag{2}$$

$$\sum_{i=1}^m x_{ij} = 1 \quad \forall 1 \leq j \leq n \tag{3}$$

$$x_{ij} \in \{0, 1\} \quad \forall 1 \leq i \leq m, 1 \leq j \leq n \tag{4}$$

where c_{ij} is the cost of assigning task j to individual i , a_{ij} is the capacity of individual i used up when task j is assigned to it, and b_i is the total available capacity of individual i . The binary variable x_{ij} is 1 if task j is assigned to individual i , and 0 otherwise. Therefore, the objective function (1) minimizes the total assignment cost; constraint (2) ensures that individual i is not assigned a number of tasks beyond its maximum capacity; constraint (3) ensures that each task is assigned to exactly one individual; and constraint (4) enforces x_{ij} to be binary. An equivalent maximum-value formulation can be readily found by interpreting c_{ij} as values rather than costs, and then changing min to max on the objective function (1). We will draw upon this formulation in Section 4.3.

While integer programming provides great flexibility for expressing discrete optimization problems, integer programs are in general NP-hard, and thus very difficult to solve (Bertsimas & Tsitsiklis, 1997). Three main categories of solution methods are exact algorithms, which are guaranteed to find an optimal solution but may take an exponential number of iterations; approximation algorithms, which provide in polynomial time a suboptimal solution as well as a bound on the level of suboptimality; and heuristics, which provide suboptimal solutions with no bound on the level of suboptimality, but have been empirically observed to find good solutions quickly.

4.1.2 Multi-objective and lexicographic optimization

Multiple criteria decision-making consists of the incorporation of more than one objective function into an optimization problem (Sayin, 2009). Formally, the multi-objective optimization problem can be expressed as

$$\begin{aligned} \max \quad & f(x) \\ \text{s.t.} \quad & x \in X \end{aligned}$$

where $X \subseteq \mathbb{R}^n$ is the feasible set and $f = (f_1, \dots, f_p): \mathbb{R}^n \rightarrow \mathbb{R}^p$, $p \geq 2$ is a vector-valued function. Here X can be continuous or discrete, expressed through constraints, and the objective function $f(x)$ can have any form.

Since the definition of optimality is no longer applicable (as each objective function can yield a different optimal solution), we are instead interested in the study of *tradeoffs* between conflicting objectives. Typically, we are interested in the set of *efficient solutions*, also called *Pareto front*, which, roughly speaking, are solutions that cannot be improved upon in at least one objective without worsening the other ones.

Closely related to multi-objective optimization is the concept of *lexicographic optimization*, in which functions are optimized according to a predefined order (Ogryczak et al., 2005). In this setting, the objective functions $f_1, \dots, f_p$ are ranked according to importance to the decision maker, where f_1 is the most important, f_2 is the next most important, and so on.

This concept captures the idea of prioritizing only one organizational preference or process duration at a time, so that when the optimal value for the first-priority objective function has been found, the other one is optimized by enforcing the first-priority objective function to take its optimal value. While this method yields an actual solution to the multi-objective optimization problem, it is unable to strike a balance between the different objective functions.

4.1.3 Volunteer management models

First, we review single-period, deterministic optimization modeling approaches, which build off of the GAP formulation (discussed in Section 4.1.1) adapting the model to each specific study. Gordon and Ekrut (2004) developed a spreadsheet-based decision support tool based on three separate models that determined daily shift schedules, volunteer assignments to each shift, and volunteer assignments to gates for the Edmonton Folk Festival. Sampson (2006) used a goal programming formulation, seeking to minimize assignment shortages and excesses for each volunteer, satisfy volunteer preferences for task assignments when possible, and minimize assignment shortages for each task. Kaspari (2010) used a two-phase model to assign volunteers. The first phase determines if a feasible assignment exists given a number of volunteers, and the second phase makes a feasible assignment of volunteers to tasks while maximizing their preferences.

Multi-period modeling methodologies were developed subsequently, combining deterministic and stochastic approaches. Garcia and Rabadi (2011) use a constraint programming and goal programming approach to manage teams in a volunteer-run church. Falasca and Zobel (2012) introduce a multi-objective optimization model assigning humanitarian volunteers to tasks across locations.

Lassiter et al. (2015) adopt a robust optimization approach to manage volunteers in humanitarian crises, accounting for task demand uncertainty. Kaur, Smith, Pazour, and Schumacher (2022) develop a deterministic optimization model aimed at volunteer retention, which accounts for organizational preferences and training time in its formulation. This model was evaluated in a stochastic simulation environment to show its performance under uncertain conditions. Ata et al. (2024) instead build on the stochastic optimization approach, exploring volunteer staffing under supply and demand uncertainty. Wu, Eftekhari, and Uichanco (2024) additionally consider the distinction between episodic and formal volunteers, as well as other factors like volunteer donations.

In this work, we develop a single-period model, since in our case study we consider episodic, one-day volunteers attending for a single shift and, furthermore, volunteers tend to perform the same tasks throughout that shift.

4.1.4 Scheduling and CP methods

We incorporate a scheduling model to accurately keep track of the process duration and account for one or more trainers overseeing the process. We refer the reader to (Pinedo, 2008) for a detailed treatment on scheduling. Of particular interest to this work is the parallel machine scheduling problem with delivery times, whose formulation we adapt from Maecker and Shen (2019). We are given a set of n jobs and m identical machines. Each job needs to be processed by exactly one machine without interruption, while each machine can handle exactly one job at a time. All jobs are available at time zero. Each job j has a specific processing time p_j and end time d_j . We consider a job-dependent delivery time q_j , which occurs immediately after completing a job on any machine. While a job is being transferred, the machine may already start processing the next job in line. Translating scheduling-related terms to our setting, the “machines” are the trainers, while the “jobs” are the volunteer subgroups that are being trained. The “delivery time” corresponds to the time that volunteers spend working after being trained. In the formulation by Maecker and Shen, the delivery time is machine-dependent, but in our case study, we model it as job-dependent (i.e., volunteer-dependent). The *tardiness* of a job T_j is defined as $T_j = \max\{C_j - d_j, 0\}$, where C_j is the time that job j is delivered. The *makespan*, defined as $C_{max} = \max_j\{C_j : 1 \leq j \leq n\}$, is the completion time of the last job to leave the system (Pinedo, 2008), while the *maximum tardiness*, defined as $T_{max} = \max_j\{T_j : 1 \leq j \leq n\}$ is the highest tardiness across all jobs in the system.

Figure 3 gives an overview of a sample training schedule for four training jobs conducted by two trainers (the “machines”), corresponding to each color (blue or red). That is, an assignment of volunteers to roles was already made, and there are four distinct roles with at least one volunteer working as part of that role. Notice that $n = 4$ and $m = 2$ in this example. We also have $p_1 = 15, p_2 = 10, p_3 = 17.5$ and $p_4 = 7.5$. Additionally, in our problem we consider that all training “jobs” have the same end time d , the time at which the setup process should end. Thus $d = d_1 = d_2 = d_3 = d_4 = 30$ in the Figure 3 example. The delivery times, that is, the times that volunteers in each role take to work after they are trained, as given by $q_1 = 12.5, q_2 = 12.5, q_3 = 12.5, q_4 = 10$. Also notice that after finishing training, a trainer can take on another training job. Hence, Trainers 1 and 2 start processing j_1 and j_3 respectively, and when one of them finishes, they may take on another training job. Hence, when Trainer 1 finishes at $t = 15$, they may start processing j_2 , while Trainer 2 is only free to process j_4 after it finishes j_3 at $t = 17.5$. Notice that the makespan is $C_{max} = 37.5$, the tardiness values are $T_1 = 0, T_2 = 7.5, T_3 = 0$, and $T_4 = 5$, which implies that the maximum tardiness is $T_{max} = 7.5$.

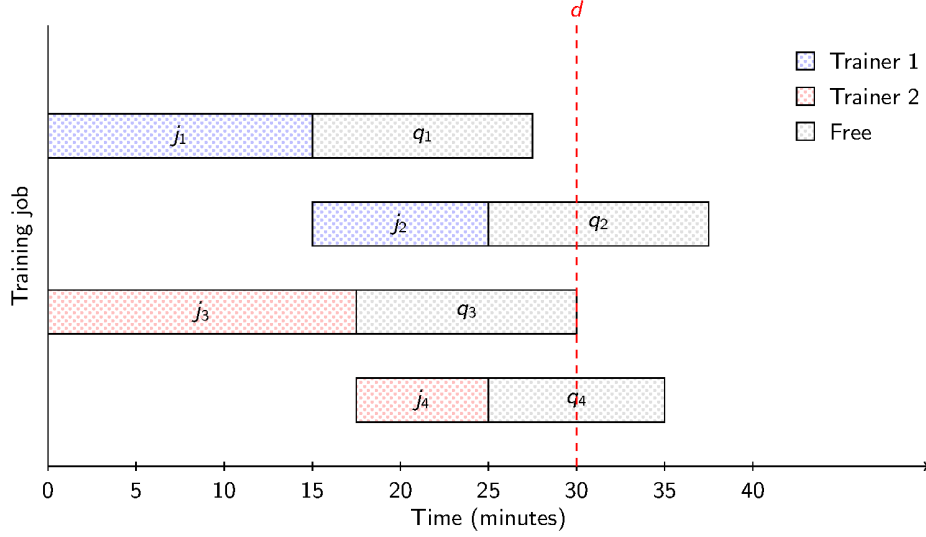


Figure 3: Scheduling model to keep track of training process duration.

Constraint programming methods are common to solve scheduling problems (Freuder & Mackworth, 2006). Of particular interest is the concept of optional interval variables introduced by Laborie and Rogerie, et al. (Laborie & Rogerie, 2008; Laborie et al., 2009), which directly model activities that may or may not be executed. An optional interval variable a is associated with four parameters; its start time $\text{start}(a)$, its end time $\text{end}(a)$, its duration $\text{duration}(a)$, and its execution status $x(a)$. The start time, end time and duration only have meaning when a is executed, in which case $x(a)$ is set to 1. If $x(a) = 0$, the start, end and duration are meaningless, and a is ignored by any constraint it is involved in. This approach has been adopted by multiple solvers, such as IBM ILOG CP Optimizer (Laborie et al., 2018) and Google OR-Tools’ CP-SAT solver (Perron & Didier, 2026).

4.2 Assumptions

We are considering a single-period model for assigning episodic, one-day volunteers from a set V to specific roles from a set J in the pantry’s floor, in a setup process that should end by time d . In our case study, we consider one “default” role, which consists mostly of minor tasks, to account for the fact that it might be too time-consuming or physically infeasible to allocate volunteers to more active roles during the setup process. That is, we do not assume that all roles are covered. As these are one-day volunteers new to the organization, they require training in order to perform their roles.

We assume that once a volunteer arrives, they are available to be trained immediately for their assigned role, though that may not happen immediately. In our case study, we consider the end time of the process the time at which all volunteers are summoned to train for the subsequent shopping process. We assume that training happens simultaneously for all volunteers assigned to the same role, and that exactly one trainer is in charge of the group while training. We assume

that volunteers start working immediately after training. Furthermore, we assume that all training and working times are independent of one another.

We assume that there is at least one regular volunteer that can serve as a trainer (at the very least, the head manager or volunteer coordinator can perform this role). We assume that as soon as a trainer finishes training, they are available to train another group, and that each trainer can only train volunteers for exactly one role at any given time. Furthermore, we assume that any trainer can train any role, that all trainers are available at the start of the process, and that every trainer has equal preference for each role to train. Once all groups are trained, we assume that working happens until the end of the process (in the real-life situation, trainers are free to supervise different groups).

We estimate the pantry’s preferences for volunteers undertaking certain roles by validating a given preference scale with the pantry managers. In our case study, we have no information about each volunteer’s preferences for specific tasks, so we assume that volunteers are indifferent regarding specific roles.

Finally, we assume that the process running late is more undesirable than finding a suboptimal volunteer-role assignment. Note that this does not necessarily mean that minimizing the makespan takes priority over finding an optimal assignment; this is only the case if the process is expected to run late.

Since we use a deterministic model, we assume that the organization can estimate all input parameters ahead of time, and these estimates will be used to create assignments and schedules.

4.3 Formulation

We model the allocation of volunteers and trainers as constraint programming (CP) models, in which we solve an assignment problem (described in Section 4.1.1) for volunteer allocation, coupled to a parallel machine scheduling problem (described in Section 4.1.4) for trainer allocation. We consider a coupled model for three main reasons. First, we want to keep track of the total process duration so that the parallel machine scheduling formulation allows us to compute the makespan and tardiness accurately. Second, the number of available trainers may be less than the number of roles to train for, and thus we need to consider that some parts of the training process happen in parallel, and some in series. Third, the GAP formulation allows us to capture organizational preferences for volunteer assignments (Kaur et al., 2022).

It is important to note that all parameters and variables are required to be integers or, in general, have a finite domain (Perron & Didier, 2026).

Our model (denoted Model 1 from this point onward) adapts its predictions depending on whether there exists an assignment that allows the process to end on time. If this is the case, the model finds an optimal assignment imposing the constraint that the process must end on time. Otherwise, it minimizes the makespan. We choose this logic to account for the fact that the process running late is more undesirable than finding a suboptimal assignment. For comparison, we formulate a model that prioritizes finding an optimal assignment and minimizing the makespan later (hereby denoted Model 2) and a model that minimizes the makespan for the given problem configuration, and then finds the maximum possible sum of assigned preferences (hereby denoted Model 3).

Unless otherwise specified, all models share the same constraints.

4.3.1 Sets

V is the set of all volunteers, indexed on v .

J is the set of all available roles, indexed on j .

M is the set of all available trainers, indexed on m .

4.3.2 Parameters

A_{vj} : the organization's preference for volunteer v being assigned to role j .

K_j : maximum allowed number of volunteers in role j .

p_j : training time for role j .

q_j : minimum desired working time for role j .

d : desired end time.

4.3.3 Variables

x_{vj} : binary variable. Equal to 1 if volunteer v is assigned to role j , 0 otherwise.

$altJ_{j,m}$: optional interval variable. If executed, models role j being trained by trainer m , with a duration of p_j . If not executed, it is ignored by all constraints.

$train_j$: optional interval variable. If executed, models role j being trained for by at least one volunteer, with a duration of p_j . If not executed, it is ignored by all constraints.

$work_j$: optional interval variable. If executed, models volunteers in role j working for a time q_j . If not executed, it is ignored by all constraints.

4.3.4 Constraints

We draw from Malapert and Nattaf (2019) for the constraint formulations.

Assignment constraints: We enforce that every volunteer is assigned to exactly one role, and that the number of volunteers on any role do not exceed its maximum capacity.

$$\begin{aligned} \sum_j x_{vj} &= 1 \quad \forall v \in V \\ \sum_v x_{vj} &\leq K_j \quad \forall j \in J \end{aligned}$$

Scheduling constraints: First, we enforce that if a role j is trained for, then exactly one trainer covers it. We achieve this via the *alternative* constraint (Laborie & Rogerie, 2008):

$$\text{alternative}(train_j, \{altJ_{j,m} \mid m \in M\})$$

That is, when $train_j$ is executed, exactly one of the variables $altJ_{j,m}$ will be executed. Furthermore, the start and end times of $train_j$ will be synchronized with the start and end times of $altJ_{j,m}$. This corresponds to trainer m being chosen to train volunteers in role j . However, if $train_j$ is not executed, then none of the variables $altJ_{j,m}$ are executed either.

Second, we enforce that no trainer can cover two or more roles simultaneously. We achieve this using the *no overlap* constraint (Laborie et al., 2009):

$$\text{noOverlap}(\{altJ_{j,m} \mid m \in M\})$$

Third, we model the available trainers as a limited resource. Observe that each role trained for “consumes” one unit of “resource”, i.e., one trainer, while the total number of trainers is $|M|$. We model this via the *cumulative* global constraint (Beldiceanu et al., 2014):

$$\text{cumulative}(\{(train_j, 1) \mid \forall j \in J\}, M)$$

Fourth, if a role j is trained for, the volunteers in such role start working immediately after finishing training. However, if it is not trained for, then the working time is ignored. We model these considerations with the `presenceOf` and `endBeforeStart` semantics (Laborie & Rogerie, 2008; Laborie et al., 2018):

$$\begin{aligned} \text{presenceOf}(train_j) &= \text{presenceOf}(work_j) \\ \text{endBeforeStart}(train_j, work_j) \end{aligned}$$

Observe that the variables $work_j$ do not require any resource, i.e., trainers, since volunteers can start working on their own.

Coupling constraints: We enforce that role j is trained for if and only if at least one volunteer is assigned to that role. We model this with the `presenceOf` semantic:

$$\text{presenceOf}(train_j) = \bigvee_v x_{vj}$$

where $\bigvee$ denotes the *logical or* operator.

Makespan modeling: Note that the makespan can be readily computed from the interval variables $work_j$ as follows:

$$C_{max} = \max_j \{\text{end}(work_j)\}.$$

4.3.5 Objective functions and solution procedures

We have two often conflicting objectives: minimizing the makespan C_{max} and maximizing the sum of assigned preferences

$$\sum_v \sum_j A_{vj} x_{vj}.$$

Following our discussion from Section 4.1.2, we cannot assume that there exists a solution that simultaneously minimizes the makespan and maximizes the sum of assigned preferences, and thus the solution to this multi-objective optimization problem consists of the Pareto front.

However, under the assumption that it is more undesirable for the process to run late than for an assignment to be suboptimal, we can propose the following solution method: if the process is expected to finish on time, maximize the sum of assigned preferences as long as the process does not run over time; otherwise, pick an assignment with minimum makespan (since the process is expected to finish late anyways). This motivates the logic for Model 1, whose solution logic is detailed in Algorithm 1. additional to the aforementioned parameters, uses a slack percentage parameter s , which attempts to force the model to find a solution whose makespan is $s\%$ times lower than the end time d . If this is not possible, it computes the true slack $s_{true} = 1 - \frac{m}{d}$ where m is the minimum makespan, and finds a solution with this slack percentage. We introduce this slack mechanic to allow Model 1 to give a makespan prediction that finishes slightly before the end time to give the process some buffer time, partially accounting for real-world situations in which some component of the process runs later than expected.

We compare the method underlying Model 1 to the following two methods. On one hand, maximizing the sum of assigned preferences and then finding the minimum makespan for this assignment (Model 2, with logic detailed in Algorithm 2); on the other hand, minimizing the makespan and then finding the maximum sum of assigned preferences for this makespan (Model 3, with logic detailed in Algorithm 3). These approaches follow the lexicographic optimization approach discussed in Section 4.1.2, but they also represent intuitive rules, prioritizing either preference matching or process duration.

4.4 Incorporation in decision support tool

We share the implementation of this model in a graphical user interface (GUI) coded with the Python library Streamlit (<https://streamlit.io>). The GUI offers basic functionality for using the model, allowing the user to introduce volunteer preference values, role information, and model preference. After the model is ran with these inputs, a volunteer-role assignment and training schedule are shown on the screen as outputs.

5 Model evaluation

5.1 Design of experiments

We design a full factorial experiment to test the performance of our models under different scenarios and showcase the tradeoff between their outputs. Table 7 details the three factors and their levels tested in the experiment, thus creating a 3×2^2 full factorial design.

Algorithm 1 Model 1 solution procedure

Solve the optimization problem

$$\min C_{max}$$

Set $m := \min C_{max}$

if $m \leq d$ **then**

$$s_{true} := 1 - \frac{m}{d}$$

$$s_{adapted} := \min\{s, s_{true}\}$$

$$d_{reduced} := \lceil (1 - s_{adapted})d \rceil$$

Solve the new optimization problem

$$\begin{aligned} \max \quad & \sum_v \sum_j A_{vj} x_{vj} \\ \text{s.t.} \quad & C_{max} \leq d_{reduced} \end{aligned}$$

else

Solve the new optimization problem

$$\begin{aligned} \max \quad & \sum_v \sum_j A_{vj} x_{vj} \\ \text{s.t.} \quad & C_{max} = m \end{aligned}$$

end if

Algorithm 2 Model 2 solution procedure

Solve the optimization problem

$$\max \sum_v \sum_j A_{vj} x_{vj}$$

Set $M := \max \sum_v \sum_j A_{vj} x_{vj}$

Solve the new optimization problem

$$\begin{aligned} \min \quad & C_{max} \\ \text{s.t.} \quad & \max \sum_v \sum_j A_{vj} x_{vj} = M \end{aligned}$$

Algorithm 3 Model 3 solution procedure

Solve the optimization problem

$$\min C_{max}$$

Set $m := \min C_{max}$

Solve the new optimization problem

$$\begin{aligned} \max \quad & \sum_v \sum_j A_{vj} x_{vj} \\ \text{s.t.} \quad & C_{max} = m \end{aligned}$$

Assignment results

Projected makespan

35 min

Projected tardiness

5 min

Volunteer	Assigned role
Volunteer 1	Role 3
Volunteer 2	Role 1
Volunteer 3	Role 2
Volunteer 4	Role 1
Volunteer 5	Role 1
Volunteer 6	Role 3
Volunteer 7	Role 2
Volunteer 8	Role 2
Volunteer 9	Role 3
Volunteer 10	Role 2



Figure 4: Sample output shown in model GUI, implemented in Streamlit

We selected these three factors based on real-life considerations. First, we consider one or two trainers that can run the training process from start to end. Second, we approximate cases in which the training and working times are close to one another (“low variance” scenario) or may be more heterogeneous (“high variance” scenario). Third, we consider cases in which the assignment options are restricted via the total maximum volunteer capacity (obtained by summing the role capacities K_j). We consider whether these numbers sum to the exact number of volunteers present (assignment options restricted), whether each role capacity is equal to the number of volunteers present (unrestricted assignment options) or whether the sum of these numbers takes a value somewhere in the middle (several assignment options).

We generate test instances for each combination of factors based on the logic that we will now specify. The parameter values and ranges, except for the preference scores, are estimated from in-person measurements at the physical pantry location from our case study. The preference scores are estimated using a simple integer scale from 1 to 10.

We fix the number of volunteers V to 15, the number of available roles J to 3, and the end time d to 30 minutes, mirroring the real-life context of our case study. We set the number of trainers to 1 or 2 depending on the current factor being tested. We generate the preference score matrix A_{vr} with integer values from 1 to 10 selected uniformly at random. Then, for a given maximum volunteer capacity V_{cap} (either 15, 30 or 45), we randomly generate an ordered triple of integers (corresponding to the three roles) whose sum is C such that each such triple is equally likely to appear (e.g., using the “stars and bars” method (Feller, 1968)). Then we generate a triple of training integer times for each role by generating three i.i.d. numbers with mean 10 from the set $\{8, 9, 10, 11, 12\}$ (“low variance” scenario) or $\{5, 6, \dots, 15\}$ (“high variance” scenario). Finally, we generate a triple of working integer times for each role by generating three i.i.d. numbers with mean 15 from the set $\{13, 14, 15, 16, 17\}$ (“low variance” scenario) or $\{10, 11, \dots, 20\}$ (“high

Factor	Levels	Description
Number of trainers $N_{trainers}$	1	One trainer available to train all volunteers
	2	Two trainers available to train all volunteers
Variance in train/work times σ^2	$\sigma = 1.41$	Low variance in training/working times
	$\sigma = 3.16$	High variance in training/working times
Maximum volunteer capacity V_{cap}	15	Low volunteer capacity; restricted assignment options
	30	Medium volunteer capacity; several assignment options
	45	High volunteer capacity; unrestricted assignment options

Table 7: Factors and levels implemented in 3×2^2 full factorial design.

variance” scenario). Observe that the “low variance” scenario corresponds to a standard deviation of 1.41, while the “high variance” scenario corresponds to a standard deviation of 3.16 in both the training and working times.

For each of the 12 combinations of factors, we generate 1000 test instances following the above logic. Then we measure the mean and standard deviation of the models’ sum of assigned preferences, makespan and maximum tardiness. For Model 1, we set the slack parameter s to 0.05.

5.2 Results

The mean values for the sum of assigned preferences, makespan and maximum tardiness in each scenario are reported in Table 9. Their standard deviations are included in Table 10. Observe that for each metric in each scenario, its mean value predicted by Model 1 lies between or is equal to one of the predicted values by Models 2 and 3. Additionally, observe that by construction of Model 1, its tardiness values are exactly equal to those of Model 3.

First, we directly compare the tradeoffs that Model 1 makes against Models 2 and 3. Table 8 shows the percentage difference in values of the sum of assigned preferences and makespan obtained from Model 1 with respect to Models 2 and 3. The “Overall” row represents the average percentage difference for each column across all combinations of factors. In particular, we may observe that Model 1’s sum of assigned preferences across scenarios is on average 8.76% higher than Model 3’s, while maintaining the same tardiness. In general, it is expected that Model 1 achieves a lower sum of assigned preferences and makespan than Model 2 (trading preference matching for less process duration), and a higher sum of assigned preferences and makespan than Model 3 (trading process duration for more preference matching).

Figure 5, Figure 6 and Figure 7 shed more light in each model’s behavior under the sum of assigned preferences metric. When $N_{trainers} = 1$, the mean sum of assigned preferences for Model 1 decreases as V_{cap} increases. However, when $N_{trainers} = 2$, the mean sum of assigned preferences increases instead when V_{cap} increases. While the respective error values indicate overlap among Model 1’s predictions across test instances with different values of V_{cap} , this suggests that Model 1 is able to shift its behavior depending on the available number of trainers. For comparison, Model 2’s mean sum of assigned preferences always increases with V_{cap} regardless of $N_{trainers}$ or σ , while Model 3’s mean sum of assigned preferences always decreases with V_{cap} instead.

			Sum of assigned preferences		Makespan	
			% Diff. w.r.t Model 2	% Diff. w.r.t Model 3	% Diff. w.r.t Model 2	% Diff. w.r.t Model 3
Overall			-7.84%	8.76%	-17.21%	7.54%
Num. trainers	Variance	Max. vol. cap.				
1	High	15	0.00%	0.00%	0.00%	0.00%
		30	-13.51%	2.95%	-30.50%	3.07%
		45	-17.42%	18.22%	-38.37%	22.95%
	Low	15	0.00%	0.00%	0.00%	0.00%
		30	-15.72%	0.22%	-30.21%	0.18%
		45	-22.39%	10.15%	-42.19%	8.09%
2	High	15	0.00%	0.00%	0.62%	0.62%
		30	-6.43%	7.03%	-12.95%	10.17%
		45	-7.19%	29.55%	-13.43%	27.08%
	Low	15	0.00%	0.00%	0.00%	0.00%
		30	-6.04%	7.05%	-20.01%	5.47%
		45	-5.34%	29.91%	-19.46%	12.81%

Table 8: Relative difference of model outputs for Model 1 compared to Models 2 and 3 for each of the 12 combinations of factors detailed in Section 5.1.

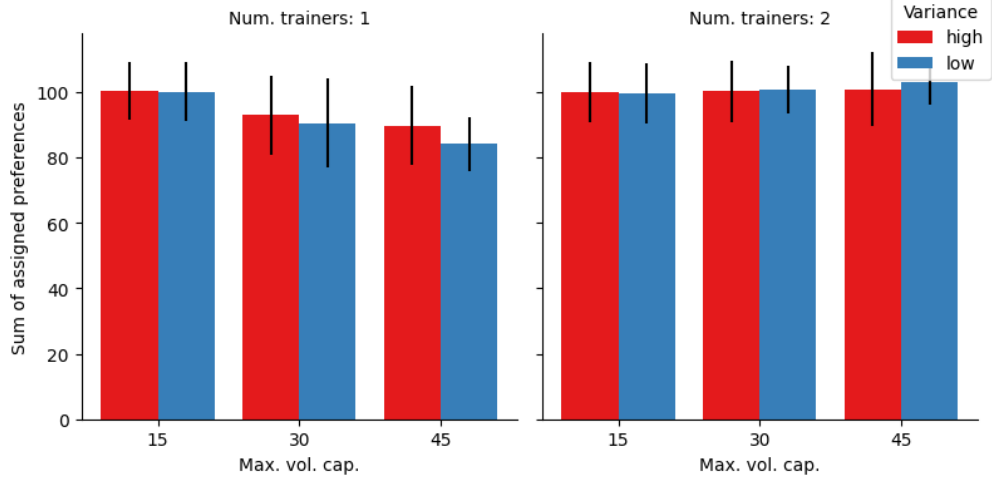


Figure 5: Mean sum of assigned preferences for Model 1 under each scenario, obtained from Table 9. Error bars correspond to the standard deviation of the sum of assigned preferences for Model 1 on each respective scenario, included in Table 10.

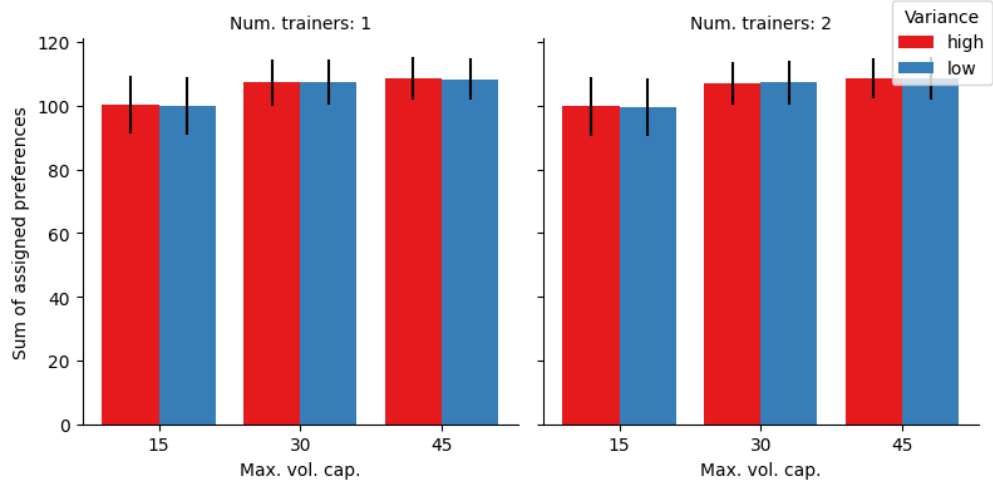


Figure 6: Mean sum of assigned preferences for Model 2 under each scenario, obtained from Table 9. Error bars correspond to the standard deviation of the sum of assigned preferences for Model 2 on each respective scenario, included in Table 10.

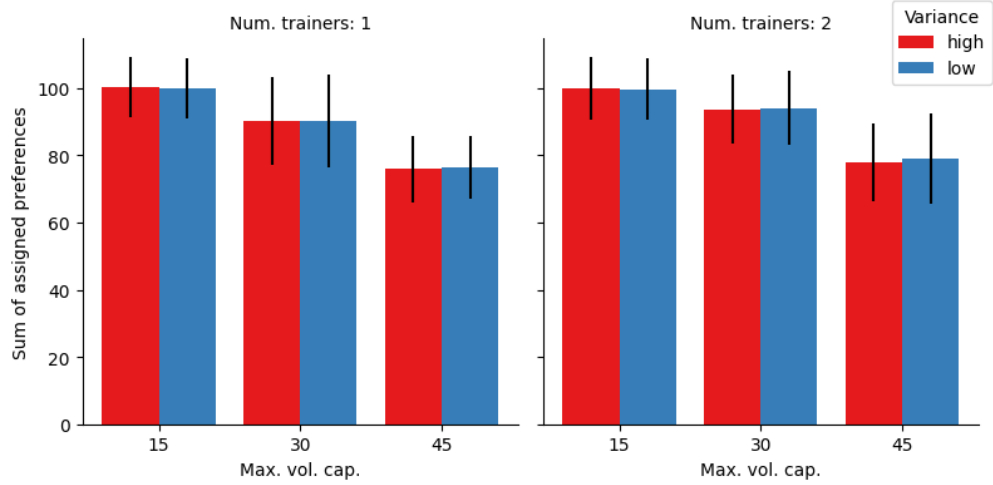


Figure 7: Mean sum of assigned preferences for Model 3 under each scenario, obtained from Table 9. Error bars correspond to the standard deviation of the sum of assigned preferences for Model 3 on each respective scenario, included in Table 10.

Num. trainers	Variance	Model Max. vol. cap.	Sum of assigned preferences			Makespan			Max. tardiness		
			Model 1	Model 2	Model 3	Model 1	Model 2	Model 3	Model 1	Model 2	Model 3
1	High	15	100.27	100.27	100.27	42.26	42.26	42.26	12.28	12.28	12.28
		30	92.81	107.31	90.14	29.35	42.23	28.47	1.37	12.27	1.37
		45	89.68	108.59	75.86	25.97	42.15	21.12	0.00	12.18	0.00
	Low	15	99.99	99.99	99.99	43.71	43.71	43.71	13.71	13.71	13.71
		30	90.43	107.29	90.23	30.53	43.75	30.48	2.03	13.75	2.03
		45	84.09	108.34	76.34	25.24	43.65	23.35	0.00	13.65	0.00
2	High	15	99.91	99.91	99.91	31.29	31.09	31.09	2.14	2.14	2.14
		30	100.17	107.06	93.59	27.06	31.09	24.56	0.06	2.11	0.06
		45	100.81	108.62	77.81	26.84	31.01	21.12	0.00	2.13	0.00
	Low	15	99.62	99.62	99.62	32.75	32.75	32.75	2.80	2.80	2.80
		30	100.76	107.24	94.12	26.28	32.86	24.92	0.00	2.93	0.00
		45	102.80	108.60	79.13	26.29	32.64	23.30	0.00	2.80	0.00

Table 9: Mean of model outputs for each of the 12 combinations of factors detailed in Section 5.1.

Num. trainers	Variance	Model Max. vol. cap.	Sum of assigned preferences			Makespan			Max. tardiness		
			Model 1	Model 2	Model 3	Model 1	Model 2	Model 3	Model 1	Model 2	Model 3
1	High	15	8.93	8.93	8.93	5.68	5.68	5.68	5.64	5.64	5.64
		30	11.97	7.24	13.02	4.22	5.98	5.12	2.53	5.89	2.53
		45	12.00	6.61	10.04	2.72	5.97	3.31	0.07	5.88	0.07
	Low	15	8.93	8.93	8.93	2.60	2.60	2.60	2.60	2.60	2.60
		30	13.59	7.11	13.65	4.09	2.83	4.16	2.18	2.83	2.18
		45	8.22	6.62	9.35	1.99	2.92	1.50	0.00	2.92	0.00
2	High	15	9.24	9.24	9.24	3.67	3.92	3.92	2.78	2.78	2.78
		30	9.45	6.83	10.28	2.24	3.84	3.39	0.40	2.67	0.40
		45	11.30	6.48	11.56	2.04	3.99	3.24	0.04	2.74	0.04
	Low	15	9.13	9.13	9.13	2.10	2.10	2.10	2.02	2.02	2.02
		30	7.34	7.01	11.05	1.65	2.11	1.51	0.00	1.95	0.00
		45	6.84	6.65	13.41	1.63	2.38	1.47	0.00	2.04	0.00

Table 10: Standard deviation of model outputs for each of the 12 combinations of factors detailed in Section 5.1.

Regarding the makespan, observe that for Models 1 and 3, the mean makespan tends to decrease as V_{cap} increases, following the increasing number of possible assignments. However, for Model 2, its mean makespan predictions tend to stay on the same level. This is consistent with the fact that Model 2 first prioritizes maximizing the sum of assigned preferences and then minimizes the makespan, ignoring time constraint considerations.

The models’ behavior on the maximum tardiness metric is similar as indicated by the makespan metric as discussed above. Note that in the $N_{trainers} = 1$ scenario, there is a sharp drop in maximum tardiness from $V_{cap} = 15$ to $V_{cap} = 30$ for Models 1 and 3, while Model 2 stays on the same level as discussed above. It is also important to observe that the maximum tardiness for all models drops from the interval (12, 14) when $N_{trainers} = 1$ to the interval (2, 3) when $N_{trainers} = 2$, stressing the difference that a single additional trainer makes in allowing the process to finish on time without sacrificing preference matching.

6 Discussion

In this work, we proposed a two-part framework to identify training components, and address space, time, and staff constraints in the setup training process for the St. James Food Pantry. We

identified relevant training components, tasks, and roles via a training needs assessment procedure. We also addressed operational challenges via a constraint programming model embedded in a decision support tool that maximizes the total assigned preference and minimizes makespan. We hope that this work serves as a groundwork for informing the design and operationalization of training programs in other food pantries and related nonprofit organizations.

There exist several limitations of the current study. First, our set of assumptions in Section 4.2 is somewhat restrictive in terms of trainer availability, throughput, skillsets, and preferences. While the training process under study is considerably fast-paced and happens over a short period of time (around 30 minutes) relative to the total shift (around 3 hours), our set of assumptions does not account for each trainer’s unique characteristics (which inevitably leads to variability in the processing time), and does not give them slack when changing between jobs. One of the most immediate improvements would be to collect data on each trainer’s preferences and ability to switch tasks, and incorporate these considerations into our multi-objective optimization formulation. Different degrees of priority may be given to each trainer characteristic to explore different kinds of assignments. Second, modern industrial-organizational psychology uses the more extensive knowledge, skills, abilities, and others (KSAO) framework, where the “others” section includes aspects like personality variables, interests, previous training, and experience (Conte & Landy, 2018). Despite not accounting in this work for characteristics falling under the “others”, previous work has dealt with motivations and attitudes of volunteers (Cnaan et al., 2022; Hustinx & Lammertyn, 2003) and its implications on volunteer management (Wu et al., 2024). Third, our formulation is single-period, so we account for only one large group of episodic volunteers, that we assume arrives all at the same time. Even though in our case study, we may have more than one group of episodic volunteers, there is typically no more than one relatively large group, and smaller groups arriving at a separate time can be handled more efficiently by the pantry staff. Multi-period formulations (such as those discussed in Section 4.1.3) should then be considered for handling the arrival of multiple groups. Fourth, task preferences from volunteers themselves are difficult to estimate in this case study, since empirical surveys to capture them were not performed. This can motivate future efforts for improvement, both implementing a mechanism for capturing volunteers’ preferences during sign-up, as well as incorporating these preferences in the model. Finally, there is inherent uncertainty on the estimation of training and working times that this model only partially accounts for via the slack parameter. This can be remedied through further development of a decision support tool to accommodate on-the-fly corrections when unexpected delays take place, or imposing further constraints to introduce more slack into the system. Alternatively, stochastic scheduling models could be considered (Pinedo, 2008) provided that adequate data for uncertainty modeling is collected.

This project produced valuable lessons into the practical considerations of conducting research into the processes of a non-profit organization. First, understanding and framing the problem at hand was the most crucial and time-consuming part of the project. A crucial aspect to the formulation was the observation that large groups of volunteers required the trainers to perform the most effort and structure the training process as much as possible, which despite being a somewhat trivial observation after the fact, it provides the foundation for a tangible problem, and is therefore the problem that this work addresses. In real operations, there are multiple process components interacting with each other, and attempting to diagnose the “root cause” (as it is often called in the process improvement literature) is difficult and time-consuming, yet it must precede any application of technical methodology. Second, communicating the project’s goals and advances to the director, receiving feedback, and adjusting expectations from the director’s and the author’s perspectives was an important part of the project. How the project is perceived by the organization may vary, and it is crucial to clearly communicate what the project aims to do to avoid misunderstandings.

Finally, projects must be tailored to each specific organization. As mentioned before, a large part of this project was framing the problem, which required careful observation of the pantry’s processes. Even if a similar problem is addressed in multiple organizations, each has their own goals, resources, and characteristics, which must be observed carefully so that project details can be tailored to each organization.

7 Acknowledgements

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8 Code availability

All code and generated data are available at <https://github.com/Leonardo-M12/Pantry-Model-SoReMo-2026>.

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A Appendix

In all tables in this section, the importance score follows the importance scale in Table 2 from 1 (not important) to 5 (crucial).

Task description	Importance
Arrange chairs, tables, shopping carts, and/or sanitizing items in the pantry's main room for client intake.	4
Unpack and place food items in designated bins, following the pantry's guidelines.	4

Table 11: Task descriptions and rated importance for the non-physically intensive setup role.

Knowledge, skills, and abilities (KSA)	Importance
Knowledge of basic sanitizing, food handling and food safety procedures.	2
Skill in moving light chairs or tables (up to 10 lb).	5
Skill in operating the pantry's shopping and storage carts.	5
Skill in opening boxes using sharp tools.	4
Skill in placing items in plastic bags or bins.	4
Ability to understand verbal and visual instructions for daily tasks.	4
Ability to communicate task progress or issues to the director or other volunteers.	4

Table 12: Knowledge, skills, and abilities (KSAs) and rated importance for the non-physically intensive setup role.

Task description	Importance
Arrange chairs, tables, shopping carts, and/or sanitizing items in the pantry's main room for client intake.	2
Load, transport, and unload item boxes/bags from storage to food stations or vice versa, using the pantry's designated carts.	4.5
Unpack and place food items in designated bins, following the pantry's guidelines.	2

Table 13: Task descriptions and rated importance for the physically intensive setup role.

Knowledge, skills, and abilities (KSA)	Importance
Knowledge of basic sanitizing, food handling and food safety procedures.	2
Knowledge of the general facility layout.	3
Knowledge of proper lifting technique.	4
Skill in lifting boxes, bags, chairs, and tables of up to 50 lb of weight.	4
Skill in operating the pantry's shopping and storage carts.	3
Skill in opening boxes using sharp tools.	4
Skill in placing items in plastic bags or bins.	2
Ability to understand verbal and visual instructions for daily tasks.	3
Ability to communicate task progress or issues to the director or other volunteers.	2.5

Table 14: Knowledge, skills, and abilities (KSAs) and rated importance for the physically intensive setup role.

Task description	Importance
Guides clients through the pantry facility to provide information about food item options.	5
Aids clients in carrying their chosen food items using the pantry's shopping carts.	5
Communicates item limit rules to clients using the facility's visual guidelines.	5

Table 15: Task descriptions and rated importance for the shopping aide role.

Knowledge, skills, and abilities (KSA)	Importance
Knowledge of the general facility layout.	3
Knowledge of the pantry's item limit guidelines.	5
Knowledge of basic sanitizing, food handling and food safety procedures.	2
Skill in operating the pantry's shopping and storage carts.	5
Skill in grabbing food items from bins.	2
Ability to recognize spoiled produce items.	3
Ability to use visual guidelines to communicate with clients speaking in a different language than the volunteer's.	4

Table 16: Knowledge, skills, and abilities (KSAs) and rated importance for the shopping aide role.

Task description	Importance
Distributes the day's meat, dairy, eggs, and/or frozen items to clients following item limit rules.	4.5
Communicates with shopping volunteers to determine appropriate item limit rules to follow for each specific client.	4
Replenishes item bins with boxes from storage provided by other volunteers.	4.5

Table 17: Task descriptions and rated importance for the distribution aide role.

Knowledge, skills, and abilities (KSA)	Importance
Knowledge of the pantry's item limit guidelines.	4.5
Knowledge of basic sanitizing, food handling and food safety procedures.	2
Skill in placing items in plastic bags or bins.	2
Skill in opening boxes using sharp tools.	3.5
Ability to understand visual guidelines for item limit rules.	4

Table 18: Task descriptions and rated importance for the distribution aide role.

Role	Knowledge, skills, and abilities (KSA)	Importance
Non-physically intensive setup	Skill in moving light chairs or tables (up to 10 lb)	5
	Skill in operating the pantry's shopping and storage carts	5
	Skill in opening boxes using sharp tools	4
	Skill in placing items in plastic bags or bins	4
	Ability to understand verbal and visual instructions for daily tasks	4
	Ability to communicate task progress or issues to the director or other volunteers	4
Physically intensive setup	Knowledge of proper lifting technique	4
	Skill in lifting boxes, bags, chairs, and tables of up to 50 lb of weight	4
	Skill in opening boxes using sharp tools	4
Shopping aide	Knowledge of the pantry's item limit guidelines	5
	Skill in operating the pantry's shopping and storage carts	5
	Ability to use visual guidelines to communicate with clients speaking in a different language than the volunteer's	4
Distribution aide	Knowledge of the pantry's item limit guidelines	4.5
	Ability to understand visual guidelines for item limit rules	4

Table 19: Relevant KSAs for each role, rated 4 or higher on importance.